

EXPERIMENT – 02

Aim: Design and train a feedforward neural network using continuous perceptron algorithm

Theory:

The **Continuous Perceptron Algorithm** is an extension of the **Discrete Perceptron**, where the output is no longer constrained to discrete values like $\{-1, 1\}$ or $\{0, 1\}$. Instead, it produces continuous outputs using a differentiable **activation function**, making it more versatile and powerful for real-world applications.

1. Model Representation

The continuous perceptron computes an output as follows:

$$y = f(w \cdot x + b)$$

where:

- $x = [x_1, x_2, \dots, x_n]$ is the input feature vector.
- $w = [w_1, w_2, \dots, w_n]$ is the weight vector.
- b is the bias term.
- $w \cdot x = \sum_{i=1}^n w_i x_i$ is the weighted sum of inputs.
- $f(\cdot)$ is a **continuous activation function**, which allows the perceptron to produce smooth outputs.

2. Activation Functions in Continuous Perceptron

Unlike the **discrete perceptron**, which uses a step function, the **continuous perceptron** uses a differentiable activation function to map inputs to outputs.

Common Activation Functions

1. Sigmoid Function (Logistic Function)

$$f(z) = \frac{1}{1 + e^{-z}}$$

- Output range: $(0, 1)$
- Used for **binary classification** problems.
- Smooth and differentiable, allowing gradient-based learning.
- **Problem:** It suffers from the **vanishing gradient problem** for large positive or negative inputs.

2. Hyperbolic Tangent (tanh) Function

$$f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

- Output range: $(-1, 1)$
- Centered around 0, making it preferable over the sigmoid function.

3. ReLU (Rectified Linear Unit)

$$f(z) = \max(0, z)$$

- Output range: $[0, \infty)$
- Used in modern deep learning models due to its efficiency.
- **Problem:** It suffers from **dead neurons** (zero gradient for negative inputs).

4. Softmax Function (For Multi-class Classification)

$$f(z_i) = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$$

- Converts logits into probability distributions.
- Commonly used in **multi-class classification**.

3. Learning Algorithm (Gradient Descent)

The continuous perceptron learns using **Gradient Descent**, unlike the discrete perceptron, which uses simple weight updates.

1. Forward Propagation

- Compute the weighted sum: $z = w \cdot x + b$
- Apply the activation function: $y = f(z)$

2. Compute the Loss

A loss function measures the difference between the predicted and actual values. Common loss functions include:

- **Mean Squared Error (MSE)** (For regression problems) $L = \frac{1}{2} \sum (y_{\text{true}} - y)^2$
- **Binary Cross-Entropy** (For binary classification) $L = -y_{\text{true}} \log(y) - (1 - y_{\text{true}}) \log(1 - y)$

3. Backpropagation (Gradient Computation)

To update the weights, we compute the gradient of the loss function with respect to weights using the **chain rule** of differentiation.

- Compute the error: $e = y_{\text{true}} - y$

- Compute the gradient of the loss w.r.t weights: $\frac{\partial L}{\partial w} = e \cdot f'(z) \cdot x$ where $f'(z)$ is the derivative of the activation function.

4. Update Rule (Gradient Descent)

- Update weights: $w \leftarrow w + \eta \cdot e \cdot f'(z) \cdot x$
- Update bias: $b \leftarrow b + \eta \cdot e \cdot f'(z)$

where:

- η is the learning rate, controlling how much the weights change at each step.

5. Repeat Until Convergence

This process is repeated until the error is minimized.

4. Properties and Advantages

Advantages

Works on non-linearly separable data

Smooth learning process due to differentiable activation functions

Can be extended into multi-layer perceptrons (MLPs)

Limitations

Slower convergence compared to discrete perceptrons

Sensitive to learning rate (too high = unstable, too low = slow learning)

Non-convex loss functions can cause convergence to local minima

5. Applications

- **Neural Networks** – Foundation for deep learning models.
- **Regression Problems** – Using linear or non-linear activations.
- **Binary and Multi-class Classification** – Used with sigmoid or softmax activations.